

Investment Strategy Challenge 2026 · Proposal for NKE Private Wealth (Chak family)

Asian Digital Transformation Snowball Note

*Growth that compounds while Asia digitalises, with capital protected
unless the basket is down more than 35% at maturity*

USD 100mn · 5-year note issued by Natixis · 9.0% p.a. snowball coupon · 65% European barrier · Trade date 17 Sep 2026

Agenda

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Client & market

Who the Chak family is, what they need, and why Asia's digital build-out now

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Risks & hedging

Investor risks, mitigants, and how the Natixis desk hedges

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Executive summary

One note that turns the family's operating expertise into compounding, protected growth

9.0%

Snowball coupon p.a.

Accrues every quarter and is paid with principal at autocall: 109% after Y1, up to 145% at Y5

-35%

Protection at maturity

European barrier: only the final level counts, so interim drawdowns cannot hurt the capital

51%

Called at first review

Risk-neutral. 83% called within 5 years; expected life 2.08 years

We recommend **a USD 100mn, 5-year Natixis snowball autocallable note** on a basket of four real, Bloomberg-listed Asian indices (Hang Seng TECH 30%, Nikkei 225 30%, TAIEX 25%, MSCI AC Asia Pacific Utilities 15%). These mirror the family's businesses in Greater China real estate, Japan technology infrastructure and renewable power. **The coupon compounds into capital rather than paying income, which matches a growth mandate. A diversified basket (not worst-of) and a maturity-only barrier make it suitable for generational wealth. Fair value is 97.3%, so issuing at par leaves Natixis 2.7% to cover hedging and margin.**

The family wants growth for the next generation, not income

Client diagnosis: third-generation dynasty, succession done, now allocating for legacy

Who they are

- Single-family office: NKE Private Wealth
- Net worth > USD 2bn; mandate USD 100mn (~5%)
- Businesses: tech infrastructure, renewable power plants, real estate in Japan and Greater China

What they want

- Capital appreciation for future generations
- Meaningful exposure to Asia's digital transformation
- To bridge traditional Asian sectors with emerging tech they know

What they can bear

- Moderate risk: preserve the legacy first
- 5-year horizon with no need for income
- Accepts limited liquidity on 5% of wealth
- Volatile backdrop: needs a buffer

Client need	Design answer in our note
Capital appreciation	Snowball coupon: 9% p.a. accrues and is paid with principal, so value compounds instead of leaking out as income
Exposure to Asian digital transformation	Four real indices mapped to the family's own industries (slide 3)
Generational preservation	European 65% barrier, diversified basket (not worst-of), BPCE-backed issuer
Volatile markets	Autocall locks in gains at the first good review date; drawdowns before maturity do not count

Why now: Asia builds the hardware of the AI economy

Investment thesis: structural growth themes, with volatility rich enough to fund a high coupon

1 AI hardware supply chain

Taiwan and Japan make the chips, equipment and servers behind global AI capex (TSMC, Tokyo Electron, Advantest).

2 Japan corporate reform

TSE pressure on capital efficiency and shareholder returns supports a re-rating of Japanese tech and industrials.

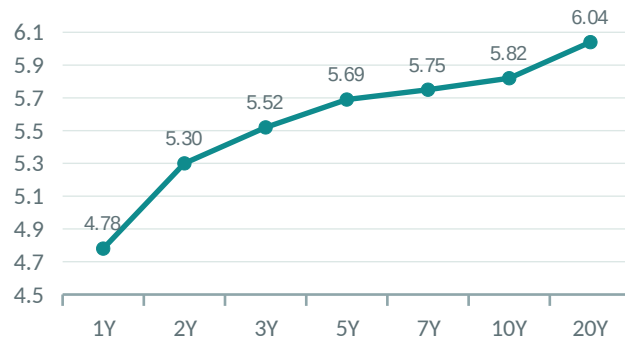
3 China platform tech reset

After the 2021–22 regulatory drawdown, Greater China internet and EV leaders trade at a discount to global peers.

4 Power for data centres

AI data centres drive grid and utility investment: steady, lower-volatility exposure that anchors the basket.

USD funding curve (case grid, %)

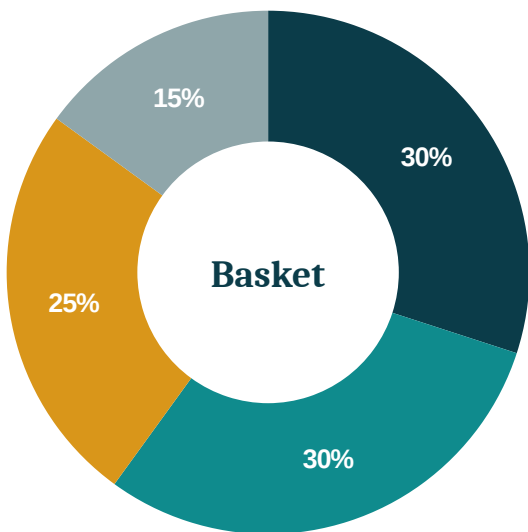


What the curve means for structuring

High USD rates make the 5Y zero-coupon bond cheap (75.8%). Combined with Asian equity volatility (basket ~19%), this funds a 9% p.a. growth coupon without a worst-of structure.

Underlying: four indices that mirror the family's businesses

Bespoke weighted basket, USD quanto (FX risk sits with Natixis); all constituents searchable on Bloomberg



- Hang Seng TECH
- Nikkei 225
- Taiwan TAIEX
- MSCI AC Asia Pacific Utilities

Index (Bloomberg)	Wt	Why it belongs (link to family)	Vol*
Hang Seng TECH HSTECH Index	30%	Greater China platform, PropTech and EV leaders: digitising the real-estate and consumer economy the family operates in	32%
Nikkei 225 NKY Index	30%	Japan tech and semiconductor equipment: home market of the family's technology infrastructure business	22%
Taiwan TAIEX TWSE Index	25%	AI supply chain and machine learning hardware (foundry, servers, networking) feeding regional logistics and retail	22%
MSCI AC Asia Pacific Utilities MXAP0UT Index	15%	Grid and power operators: the smart-grid layer above the family's renewable plants; lowers basket volatility	14%

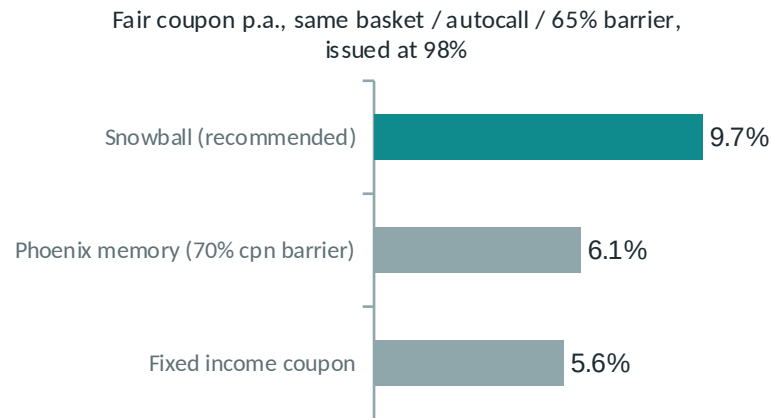
Why a weighted basket, not worst-of: one index collapsing (e.g. a China regulatory shock) is cushioned by the other three. Basket vol is ~19% vs 32% for HSTECH alone, and correlation risk stays with the desk.

*Assumed 5Y implied vols for pricing. The draft's thematic indices (e.g. 'Greater China PropTech & NLP') are not Bloomberg-listed, so we map each theme to a real index.

Term sheet: 5-year snowball autocall, 65% European barrier

Why snowball: same risk as an income autocall, but a much higher coupon and true compounding

Issuer	Natixis SA (BPCE Group)
Notional / currency	USD 100,000,000 · quanto USD
Trade / maturity	17 Sep 2026 / 17 Sep 2031 (5Y)
Underlying	Weighted basket of 4 indices (slide 3)
Observation	Quarterly, autocall from Q4 (Year 1)
Autocall trigger	Basket $\geq$ 100% of initial level
Snowball coupon	9.0% p.a. × years elapsed, paid at call
Barrier	65% European (maturity only)
Below barrier	Principal × final basket level
Issue price / fair value	100% / 97.3%

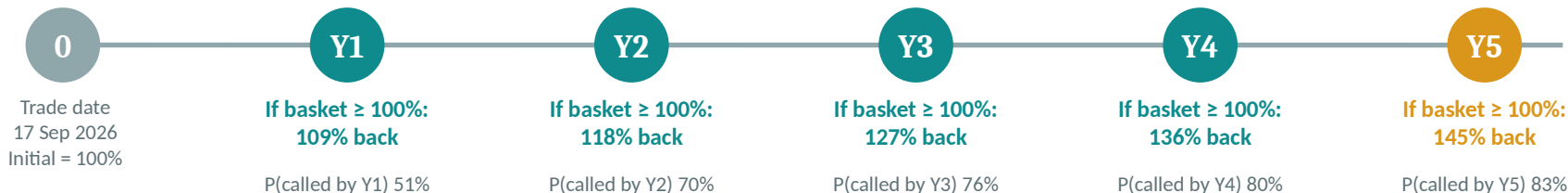


Design call: A quarterly-paid coupon is only worth ~5.6% at the 5.69% funding rate, so the original 8.5% fixed-coupon draft would cost Natixis 5.1% above a 98% issue price. Paying the coupon only at call lifts it to 9.7% fair. We offer 9.0% and keep the rest as hedging margin.

How it pays: one question each quarter, one at maturity

Observation timeline (quarterly from Year 1; one example review date per year shown)

Quarterly checks between dates pay pro-rata (e.g. Q6 = 113.5%)



A Autocalled

Basket $\geq 100\%$ on any review from Q4. Note ends with $100\% + 9.0\% \times \text{years}$.

83% risk-neutral probability

B Not called, final $\geq 65\%$

Basket ended between 65% and 100% at Year 5. Full principal is returned, with no coupon.

10% risk-neutral probability

C Not called, final $< 65\%$

Barrier breached at maturity. Principal $\times$ final level, e.g. basket at 55% returns 55%.

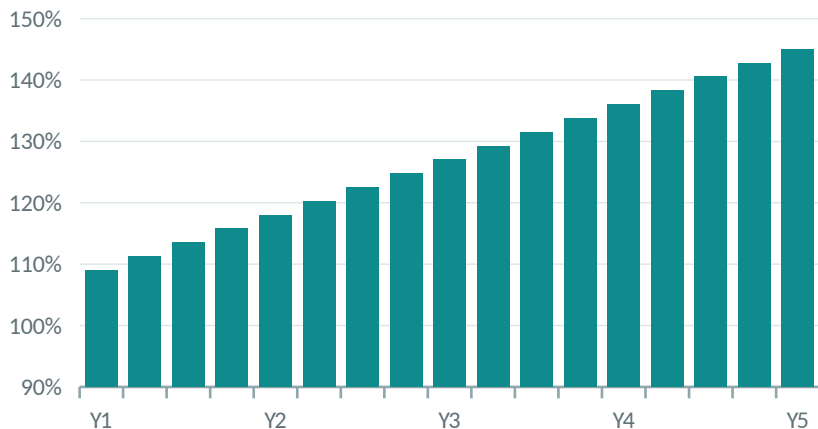
7.7% risk-neutral probability

Probabilities are risk-neutral (drift = funding rate – dividends). Real-world outcomes under different equity returns are on slide 9.

Payoff diagrams: what the family receives, and when

Left: redemption if called at each review date. Right: maturity payoff if never called, vs. owning the basket directly

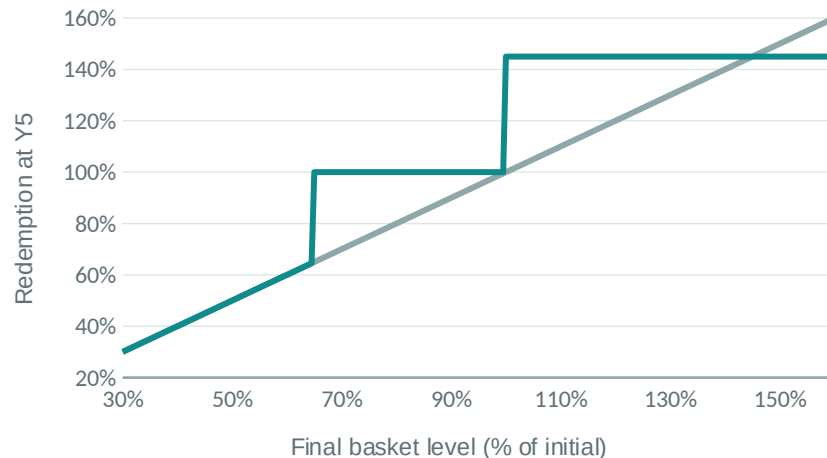
Redemption at autocall (% of notional), by review quarter Q4-Q20



Y1: 109% (USD 109mn)
Y2: 118% (USD 118mn)
Y3: 127% (USD 127mn)
Y4: 136% (USD 136mn)
Y5: 145% (USD 145mn)

Coupon grows linearly; there is no compounding cap until maturity. The longer the wait, the bigger the payout.

Maturity payoff (note not called earlier)



— Snowball note — Basket directly

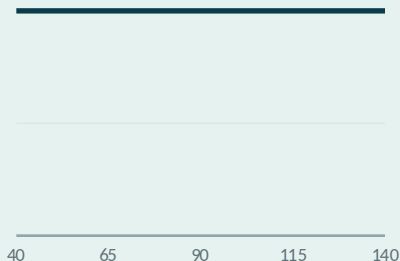
- < 65%: 1-for-1 loss, same as the basket
- 65-100%: 100% back (the buffer zone)
- ≥ 100%: 145% (Y5 call)

Under the hood: a bond plus three option positions

Replication used by the Natixis desk to price and hedge (values in % of notional, at trade date)

Zero-coupon bond

Natixis 5Y funding at 5.69%



+

Long autocall digitals

Pay 9% × years the first time the basket $\geq 100\%$ on a review date



+

Short European put

Strike 65%, killed if the note autocalls: the client takes losses below 65%



+

Short digital put

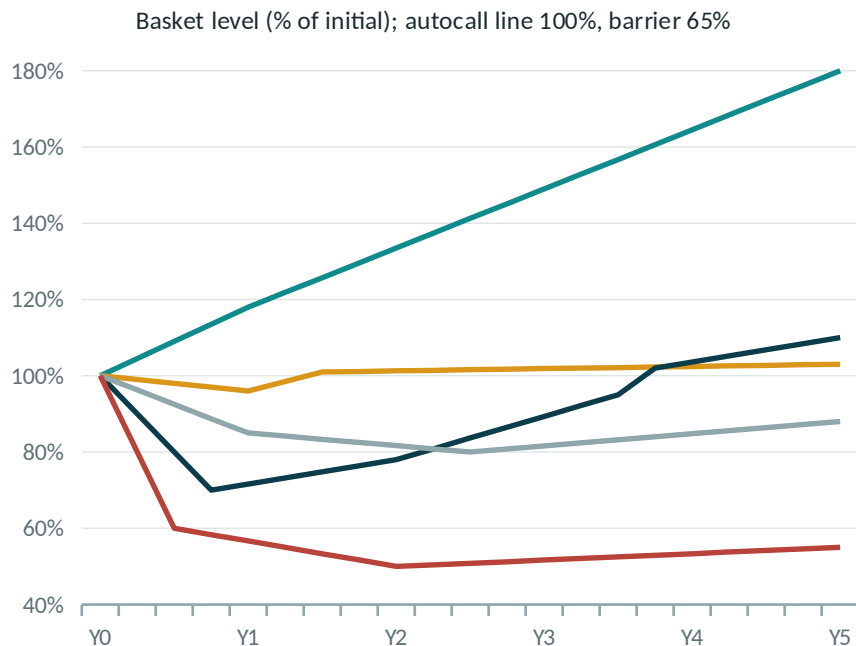
Pays 35% if final < 65%: the jump from 100% to 65% at the barrier



Payoff at Y5 = bond + digitals – put – digital put. Fair value 97.3% vs 100% issue price → 2.7% for hedging costs and Natixis margin. The premium the client earns by selling the 65% downside (the two red legs) is what funds the 9% snowball coupon.

Scenario analysis: five paths for Asian markets

Hypothetical quarterly basket paths and what the USD 100mn note returns in each



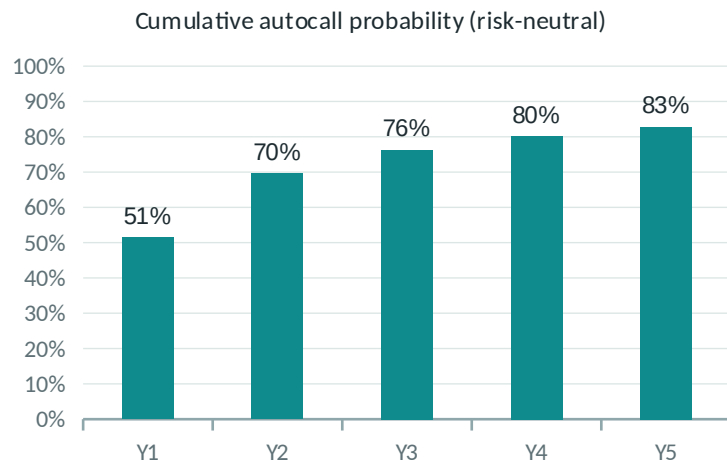
Scenario	Outcome	Payout	USD mn	IRR
AI capex boom	Called Q4 (1y)	109.0%	109.0	9.0%
Range-bound Asia	Called Q6 (1.5y)	113.5%	113.5	8.8%
2022-style tech drawdown, slow recovery	Called Q15 (3.75y)	133.8%	133.8	8.1%
Lost decade (sideways below par)	Par at Y5	100.0%	100.0	0.0%
Geopolitical shock, no recovery	Barrier hit at Y5	55.0%	55.0	-11.3%

Reading: the note earns ~8–9% p.a. in up, flat and recovering markets, including a 30% interim drawdown, because only the final level is tested against the barrier. It returns par in a lost decade, and loses like the basket only if Asia ends 5 years down by more than 35%.

Paths are illustrative and not forecasts. Historical rolling-window back-test to be run on Bloomberg index history (method in appendix).

Simulation back-test: most paths end early, at a profit

Monte Carlo outcome distribution: risk-neutral and under three real-world equity return assumptions



Metric	Risk-neutral	0% equity p.a.	4% equity p.a.	8% equity p.a.
Called within 5Y	83%	69%	80%	89%
Expected life (yrs)	2.1	2.6	2.2	1.8
Mean IRR	6.4%	3.5%	5.9%	7.4%
Par, no coupon	10%	12%	10%	7%
Capital loss	7.7%	19.6%	9.9%	4.0%
Avg loss if loss	47%	50%	48%	46%
ES 95% (loss)	51%	63%	55%	36%

51%

called at the first review
(109%)

92%

of paths return at least
par

3.6%

expected capital loss
(prob × severity)

Risks for the family, and how Natixis hedges its side

Transparent disclosure plus a desk hedging plan for every Greek

Risk	What could happen	Mitigant
Market / barrier	Below 65% at Y5, loss is 1-for-1 from 100%	Diversified basket; maturity-only barrier; 35% buffer
Opportunity cost	Called early means reinvestment risk; flat 65–100% means 0% for 5Y	Snowball pays more the longer it waits; roll-over at call
Capped upside	Basket +80% still pays 145% at most	15% of family wealth stays in direct equity
Issuer credit	Unsecured Natixis / BPCE exposure	BPCE senior rating; optional collateralised wrapper
Liquidity	Bespoke note; secondary price at Natixis bid	Natixis daily indicative price; 5% of net worth only

Pricing sensitivity (fair coupon): base 9.7% · vol ± 3 pts 8.5%–11.0% · corr ± 0.15 8.8%–10.4%. 9.0% is fundable unless vols fall ~ 3 pts before trade date.

Delta / vega

Natixis is long the 65% put the client sells (long downside vega). Delta-hedge with HSTECH / NKY / TAIEX futures; sell listed OTM index puts to recycle vega and skew

Correlation

Long a basket put means long correlation. Offset by buying worst-of puts or a dispersion trade (short basket vol, long single-index vol)

Quanto FX

USD payout on JPY/HKD/TWD indices: FX forwards and quanto adjustment in price (FX-equity correlation)

Rates & funding

Fixed snowball accrual hedged with USD swaps; proceeds fund the 5.69% ZCB

Appendix A: pricing assumptions and methodology

Inputs to refresh on Bloomberg at trade date (BVOL / OVDV for vols, CORR for correlations)

Index	Weight	Vol	Div	HSTECH	NKY	TWSE	MXAP0UT
HSTECH	30%	32%	1.0%	1.00	0.45	0.55	0.35
NKY	30%	22%	1.8%	0.45	1.00	0.60	0.35
TWSE	25%	22%	2.8%	0.55	0.60	1.00	0.35
MXAP0UT	15%	14%	3.5%	0.35	0.35	0.35	1.00

Model. Correlated GBM for each index, 200,000 paths, quarterly steps. Risk-neutral drift = USD funding (linear interpolation of case grid) – dividend yield. Quanto adjustment ignored (assumed hedged by the desk; to be added with FX-equity correlations).

Discounting. USD funding grid from the case rules; 5Y discount factor 0.758.

Coupon solve. Bisection on coupon so that PV = 98% (2% hedging cost and margin).

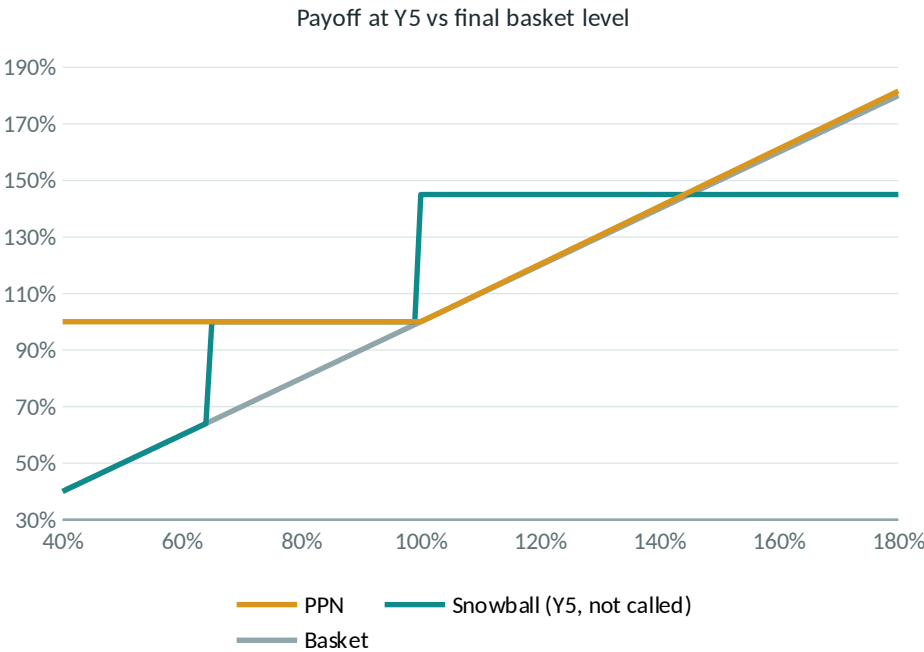
Historical back-test (to run). Monthly Bloomberg history since 2015 for all four indices. Launch a hypothetical note every month, apply the same autocall and barrier rules, and record payout, life and IRR. Report hit rates by launch year.

Design variant	Fair cpn
Snowball, 70% barrier	10.1%
Snowball, 65% (base)	9.7%
Snowball, 60% barrier	9.2%
Step-down trigger –5%/yr	7.6%
First call at Y2	10.0%
Fixed coupon, 70% barrier	5.9%
Phoenix memory, 70% cpn barrier	6.1%

All variants priced at 98% with the same basket and inputs. Code: `pricing/scenario2_autocall_mc.py`

Appendix B: alternative for a risk-averse patriarch

100% principal-protected note on the same basket



PPN term	Value
Zero-coupon bond (5Y)	75.8%
Option budget (at 98%)	22.2%
5Y ATM basket call	21.7%
Upside participation	102%
Minimum redemption	100%

Trade-off: High USD rates fund ~102% participation with full capital protection, but nothing is earned unless the basket ends above 100% after 5 years, and there is no early exit. The snowball earns 9% p.a. in flat-to-up markets and pays out early. We recommend the snowball, and keep the PPN as the fallback if the investment committee requires hard protection.